A Network and Its Reflection: What True, False, and Noise Self-Model Broadcasts Do and Do Not Change in an Evolving Multi-Agent Network

Ivan Bhargava and Harsh Modi

IIIT-Bangalore
ivan.bhargava@iiitb.ac.in, harsh.modi@iiitb.ac.in
WWW home page: https://github.com/Ivan825/Project-ONE

Abstract. Networks often act on measurements of their own state, a loop in which descriptions can change what they describe. We build a controlled evolving-network model to ask when this loop causes genuine self-fulfilling effects. Across 3,127 deterministic runs with paired-seed comparisons, measurement without broadcast changes nothing. A seemingly strong "pull" toward false and replayed self-models disappears when untreated twin trajectories are scored against matched references, showing how intrinsic dynamics mimic performativity. Broadcasts do have real effects: they reshape cooperation and network structure, and whether a false self-model comes true, in the outcome sense, depends on how agents respond. Finally, attention to the broadcast is heritable and evolves downward under consequential feedback — most steeply where that feedback carries high corrective drive — surviving three architectural controls and an ecological ensemble. The results separate apparent performativity from genuine feedback effects and show that reliance on a collective self-model can itself evolve.

Keywords: adaptive networks, agent-based modeling, reflexivity, self-fulfilling dynamics, performative prediction, evolution of trust

1 Introduction

Adaptive networks — graphs whose topology coevolves with the dynamical states of their nodes — are a central object of network science [1, 2]. A distinctive subclass has received far less controlled attention: networks that respond to *descriptions of themselves.* Financial markets move on published indices computed from their own trades [3]; universities restructure around rankings derived from their own behavior [4]; recommendation platforms amplify items *because* an internal measurement declared them trending. In each case a macroscopic measurement is compressed, published, and re-enters as an input to microscopic decisions — the loop Merton called the self-fulfilling prophecy [5], whose measure-distorting variant is Goodhart's law [6].

Empirical study of this loop is confounded: one can rarely withhold the measurement, falsify it in controlled directions, or replay history under identical randomness. We therefore built a minimal simulated world where the loop

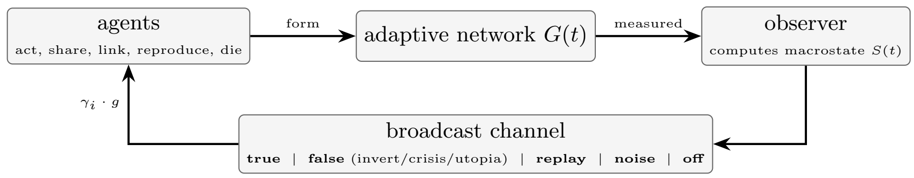

Fig. 1. The reflexive loop and its corruption points. Local actions generate the network; the observer compresses it into a self-model; the broadcast channel returns it truthfully, distorted, replayed, as noise, or not at all. Action-propensity responses scale with heritable sensitivity γ_i and gain g; the hub-targeted pruning probability scales with γ_i alone.

can be opened, closed, and corrupted at will, every run is exactly reproducible from (configuration, seed), and agents are transparent trait vectors rather than learned policies, so every pathway from broadcast to response is inspectable.

Our contributions are: (1) an open, deterministic experimental platform coupling an evolving agent ecology, an adaptive interaction network, a global observer, and a controllable broadcast channel; (2) a six-condition causal design — verifying observer-side-effect invariance and isolating the effect of broadcasting, with matched-range noise and replayed-self-model controls — plus a *passive-counterfactual methodology* for reflexivity metrics, under which a natural "story pull" statistic is fully accounted for by untreated twin trajectories; (3) evidence for what broadcasts *do* change: truthful feedback doubles costly cooperation, distorted broadcasts mobilize or pacify depending on the lie–response pairing (reversing under conformity), and false and noise broadcasts densify the network while also reducing hub concentration; and (4) receiver sensitivity as a continuous heritable trait, whose *population distribution shifts systematically under feedback*, tracking the broadcast-implied corrective-drive intensity and surviving three architectural controls and a randomized ecological ensemble — reliance on the observer is not updated by rule; it evolves. Figure 1 summarizes the design.

2 Related Work

Adaptive and temporal networks. Coevolution of state and topology [1, 2] covers our setting; robustness to targeted node removal is classical [7], and we use hub removal as a standardized perturbation. **Global coupling.** A mean field fed back to units reshapes collective dynamics [8]; ours is multidimensional, potentially *false*, and its units are born, die, and evolve. **Reflexivity and reactive measurement.** Self-fulfilling prophecy [5], reflexivity in markets [3], and the reactivity of rankings and metrics [4, 6] motivate our false-broadcast conditions; here they are dosable and ablatable. **Performative prediction and endogenous signals.** Predictions that alter the distribution they forecast are formalized as performative prediction [19, 23], including networked settings [20] and populations deciding whether to trust predictions of collective outcomes [21]. There trust is *updated* from past accuracy; here attention is inherited and mutated with no rule referencing reliability. Self-fulfilling signals of an endogenous network state arise in congestion games [24], and causal identification of performative effects without randomization is an active problem [22]. Ours is narrower and complementary: a concrete false-positive failure mode of directional-alignment metrics, diagnosed with paired passive twins (Sect. 5.2), plus turnover and a heritable receiver-sensitivity trait absent from these models. **Signaling and the evolution of trust.** In sender–receiver games receivers can evolve to ignore unreliable signals [9]; here the signal is *self-referential* and attention to it is a continuous heritable trait in an energy economy, not a payoff matrix. **Agent-based modeling.** Collective opinion and cooperation are mainstays [16, 17]; we follow the generative ABM tradition [10, 11], with an engine using ideas from Mesa [12] and NetworkX [13]. Multi-agent RL treats global state as an *architectural* aid [14]; we treat it as an *experimental variable* with controlled veracity. For language-model agent populations [15], this rule-based setting provides causally clean baselines for the shared-dashboard designs natural there.

3 The Model

3.1 Agents and ecology

Each agent i carries a heritable trait vector $\theta_i = (\mathrm{coop}, \mathrm{expl}, \mathrm{risk}, \mathrm{soc}, \mathrm{share}, \gamma_i) \in [0,1]^6$, where γ_i is its *global sensitivity* — how strongly broadcasts modulate its behavior — and $\bar{\gamma}$ its population mean. Agents hold energy; each step they pay metabolism and choose one action — harvest, share (costly transfer to a neighbor), connect, prune, reproduce, or idle — with probabilities proportional to trait- and state-dependent propensities. The resource pool regenerates logistically; reproduction requires an energy threshold and copies traits with Gaussian mutation ($\sigma = 0.05$); energy exhaustion or age causes death; founding ages are staggered to avoid cohort artifacts.

3.2 Observer and broadcast conditions

Every $\Delta t = 10$ steps the observer computes the macrostate $S(t)$: population, fragmentation (1 minus largest-component share), degree- and betweenness-concentration, Freeman centralization [18], per-capita costly-helping rate, energy Gini, turnover, and mean traits (logged, never broadcast). The broadcast $b(t)$ carries five of these, normalized to $[0, 1]$: *fragmentation*, *degree concentration* (top-5% degree share, distinct from Freeman centralization), *cooperation*, *inequality* and *turnover*. Six conditions define what agents receive (Table 2). Distorted broadcasts come in three modes: *invert* ($b_k = 1 - s_k$ componentwise), *crisis* (fixed alarming values: fragmentation 0.9, degree concentration 0.9, cooperation 0.1, inequality 0.9, turnover 0.9), and *utopia* (reassuring: 0.05, 0.2, 0.9, 0.1, 0.2).

3.3 Response model

Each step, agent i selects one action a with probability proportional to

$$w_i(a,t) \;=\; w_a\big(\theta_i, x_i(t)\big) \;+\; \gamma_i\, g\, f_a\big(b(t)\big), \tag{1}$$

Table 1. Model parameters (defaults; campaign overrides noted in the text).

initial population	150	resource capacity K	6000	observer interval Δt	10 steps		
population cap	1200	base regeneration	60/step	feedback gain g	0.8 (swept 0.2–1.6)		
initial edges/agent	3 (random)	regeneration rate r_{reg}	0.12	mutation σ	0.05		
initial energy	20	harvest cap	2.5/step	initial traits	mean 0.5, s.d. 0.15		
metabolism	0.8/step	connect cost / share	0.25 / 3.0	reprod. threshold / cost	30 / 16		
run length	3000–4000 steps	min. reproduction age	15 steps	lifespan (mean $\pm$ s.d.)	400 ± 80 steps		
shock time t_{shock}	2000 (1500 in the 2× scale check)						

Table 2. Experimental conditions; all start from matched parameter distributions with paired seeds.

Condition	Observer	Agents receive	Role
A — Local only	off	nothing	baseline
B — Observed, blind	on	nothing	measurement control
C — True feedback	on	accurate $S(t)$	main treatment
F — False feedback	on	distorted $S(t)$	reflexivity probe
N — Noise feedback	on	uniform random, matched ranges	stimulation control
R — Replayed self-model	on	another run's genuine $S(t)$	structure control

where $x_i(t)$ is i's local state, g the global feedback gain, f_a the response rule, and the base propensities are harvest $1 + \mathbf{1}[e_i < 15]$, share $0.6\theta_{\text{coop}}\theta_{\text{share}}$, connect $0.4\theta_{\text{soc}} + 0.3\theta_{\text{expl}}$, prune 0.08, reproduce $0.5\,\mathbf{1}[e_i \geq 30]$ and idle 0.2 (e_i = energy). In the primary *corrective* regime agents repair reported deficits: $f_{\text{connect}} = \max(0,\, b_{\text{frag}} - 0.3)$, $f_{\text{share}} = \max(0,\, 0.5 - b_{\text{coop}}) + 0.5\max(0,\, b_{\text{ineq}} - 0.5)$, $f_{\text{harvest}} = \max(0,\, b_{\text{turn}} - 0.5)$, and $f_{\text{prune}} = 0.3\max(0,\, b_{\text{cent}} - 0.6)$. In the *conformist* regime agents imitate the reported norm instead: $f_{\text{share}} = b_{\text{coop}}$, $f_{\text{prune}} = 0.5\, b_{\text{frag}}$, $f_{\text{connect}} = \max(0,\, 0.6 - b_{\text{frag}})$, and $f_{\text{harvest}} = b_{\text{ineq}}$. Two mechanisms act on *target selection* rather than action choice, in both regimes: while reported degree concentration exceeds 0.6 a pruning agent cuts its highest-degree neighbor (ties by id) with probability γ_i — g scales action propensities only, not this probability — else a uniform neighbor; and new links attach to a neighbor-of-a-neighbor with probability 0.7 when available, else a uniform non-neighbor — the triadic closure behind Sect. 3.4. All rule constants are fixed across experiments (Table 1).

3.4 Emergent network structure

The emergent networks are dense, clustered, and heterogeneous: at $t = 2{,}000$ the giant component (six graphs: A and C × 3 seeds) has mean path length $L \approx 1.8$–2.5, clustering 1.8–2.9× a matched Erdős–Rényi null, and degree coefficient of variation 0.6–0.74; against the stricter degree-preserving (configuration-model) null most of the clustering excess traces to the degree sequence (ratios 1.0–1.6), so we make no small-world claim. Feedback acts on a small, dense, degree-heterogeneous clustered graph, not a homogeneous mixing population.

3.5 Reproducibility

The simulation is deterministic given (configuration, seed): one PRNG drives all behavioral choices and a separate dedicated stream generates stochastic broadcast signals, so constructing the noise signal never advances the behavioral PRNG; agents act in sorted order and a state hash certifies replay (identical hashes verified across machines and operating systems).

4 Experimental Design

The standardized shock removes, in a single step at $t_{\text{shock}} = 2000$ (except in the 2× scale check: $t_{\text{shock}} = 1500$ in a 3,000-step run), the highest-degree living agents (ties broken by agent id): a fixed 10 hubs in the mild variant, the top $\max(1, \lfloor 0.4\,|V_{\text{live}}| \rfloor)$ in the harsh variant. Four outcome *families* were fixed before the first campaign: (1) post-shock recovery time — the first $t > t_{\text{shock}}$ at which the living population regains 90% of its pre-shock baseline, defined as its mean over $[t_{\text{shock}}-500,\, t_{\text{shock}})$; (2) mean post-shock fragmentation; (3) per-capita costly-helping rate, averaged after a 500-step burn-in; (4) self-model correspondence, operationalized as *story pull*: with $s_k(t)$ the value of macrostate component k at observer tick t, $b_k(t)$ the broadcast value, and $\varepsilon = 0.02$,

$$P \;=\; \Big\langle\, \operatorname{sign}\big(b_k(t) - s_k(t)\big)\,\big(s_k(t{+}1) - s_k(t)\big) \,\Big\rangle_{(t,k)\,:\,|b_k(t)-s_k(t)|>\varepsilon}, \qquad (2)$$

averaged over all qualifying (tick, component) pairs across the five broadcast components; $P > 0$ means the macrostate moves toward the reference where the two differ. The threshold ε makes P well defined under false and noise broadcasts (and undefined for C, whose broadcast equals the current state).

Outcome pre-specification. Eq. 2 is a *post hoc* directional refinement of the fourth family, adopted because the original distance-based form is degenerate under truthful feedback; both yield the same raw F-vs-N ordering. No pre-specified outcome was dropped, and all other reported measures — including the evolved trait change — are exploratory.

Passive counterfactual. P can also be evaluated on trajectories that never received the reference: for each treated run we compute P on the paired-seed *untreated* trajectory (A/B) scored against a matched reference of the same type, defining the *counterfactual performativity gap* $\Delta P = P_{\text{actual}} - P_{\text{passive}}$ (Sect. 5.2). Campaigns: a *flagship* run (conditions A/B/C/F_{invert}/N × 50 paired seeds, mild shock), a *harsh-shock* replication (40% hub removal, ×30 seeds), a *sensitivity sweep* (feedback gain $g \in \{0.2, 0.8, 1.6\}$ × {A, C, F_{invert}, F_{crisis}, F_{utopia}, N} × 20 seeds), a 2× *scale check* (doubled population and resources), a conformist *response-regime campaign* (15 seeds per cell), and a *replay control* (30 receiving runs on the harsh-shock seed list plus 15 auxiliary source runs): the core campaigns comprise $250 + 150 + 360 + 32 + 90 + 30 + 15 = 927$ runs of 3,000–4,000

Table 3. Main paired comparisons (Wilcoxon signed-rank on per-seed differences; matched-pairs rank-biserial r; bootstrap 95% CI of the median difference). FL = flagship (50 pairs), HS = harsh shock (30 pairs). "adj." compares $\Delta P = P_{\text{actual}} - P_{\text{passive}}$ between conditions (Sect. 5.2). Multiplicity: over the eight rows carrying a point p, Benjamini–Hochberg and the stricter Holm–Bonferroni both leave every decision unchanged. Enlarging the family to all 80 paired contrasts in the paper, BH still leaves every decision here unchanged; the stricter Holm demotes one, the weakest ($p = 9.5 \times 10^{-4}$). Other demotions fall in exploratory checks. Exploratory analyses are read through effect size and replication rather than adjusted significance.

Outcome	Comparison	Δmedian [CI]	r	p
all state outcomes (FL, HS)	B vs A	all differences $\equiv 0$	—	—
story pull, raw (HS)	F vs N	$+0.0066$ $[0.0037, 0.0095]$	0.89	1.7×10^{-6}
story pull, adj. (HS)	F vs N	$+0.0022$ $[-0.0003, 0.0034]$	0.36	0.084
story pull, adj. (FL)	F vs N	$+0.0004$ $[-0.0016, 0.0022]$	0.08	0.63
cooperation (FL)	C vs A	$+0.200$ $[0.172, 0.239]$	1.00	8.9×10^{-15}
cooperation (HS)	C vs A	$+0.209$ $[0.169, 0.248]$	0.99	9.3×10^{-9}
fragmentation (FL)	F vs A	-0.017 $[-0.034, -0.006]$	-0.52	9.5×10^{-4}
$\Delta\bar{\gamma}$ (HS)	F vs A	-0.336 $[-0.456, -0.281]$	-1.00	1.9×10^{-9}
$\Delta\bar{\gamma}$ (HS)	N vs A	-0.323 $[-0.431, -0.240]$	-0.94	2.0×10^{-7}
recovery time (FL, HS)	C, F, N vs A	Δmedian $= 0$	$\lvert r\rvert < 0.5$	> 0.22

steps; the three architectural controls (Sect. 5.5) add 300 + 300 + 400 and the ecological ensemble (Sect. 5.7) 1,200, for 3,127 in total. Because conditions share seed lists the design is paired; the primary analysis is a Wilcoxon signed-rank test on per-seed differences with matched-pairs rank-biserial r and a bootstrap 95% CI on the median difference. Unpaired Mann–Whitney U with Cliff's δ is a secondary robustness view; all inferential claims follow the paired analysis, and other logged measures are exploratory. Per-run distributions for every reported outcome ship with the released data; cells carry 15–100 paired seeds; the smaller checks — 2× scale ($n = 8$), the ecological ensemble (10 per ecology) and the structural nulls (six graphs) — are flagged where they appear.

5 Results

Table 3 summarizes the main paired comparisons; the following subsections unpack them.

5.1 Measurement alone is causally inert

Under paired seeds, condition B is trajectory-identical to A: every per-seed difference is exactly zero, on every outcome, in both campaigns. The instrument acquires causal power only when it broadcasts.

5.2 Apparent story pull is explained by a passive counterfactual

Except where stated, results concern the corrective regime; Sect. 5.4 shows how they change under conformity. Evaluated naively, the story-pull statistic appears to show strong self-fulfilling steering. Raw P (harsh shock; medians with bootstrap 95% CIs) is positive in all three non-truth conditions: F 0.0078 $[0.0066, 0.0100]$ and N 0.0014 $[0.0001, 0.0023]$. It is largest for a sibling world's replayed genuine self-model, condition R, at 0.0241 $[0.0197, 0.0265]$ — 30 receivers on the harsh-shock seed list, one of 15 sources each. The hierarchy R > F > N is reliable: F vs N gives $r = 0.89$, $p \approx 1.7 \times 10^{-6}$, and it survives both one receiver per source ($n = 15$, $r \geq 0.98$) and a source-clustered bootstrap.

A short observation shows why raw P can mislead. Under the inverted reference $b_t = 1 - s_t$ we have $\mathrm{sign}(b_t - s_t) = \mathrm{sign}(\frac{1}{2} - s_t)$. Suppose an untreated component simply mean-reverts toward mid-range: $s_{t+1} - s_t = -\kappa(s_t - \frac{1}{2}) + \eta_t$ with $\kappa > 0$ and $\mathbb{E}[\eta_t \mid s_t] = 0$. Then $\mathbb{E}[\mathrm{sign}(b_t - s_t)(s_{t+1} - s_t) \mid s_t] = \kappa \, |s_t - \frac{1}{2}| > 0$: ordinary mean reversion masquerades as attraction toward the mirror, with zero causal feedback. Post-shock relaxation plays the same role for the replayed reference, which tracks where recovering trajectories go anyway. The remedy is the *counterfactual performativity gap* ΔP. Score the paired-seed *untreated* trajectory (A, or the identical B) post hoc against a matched reference of the same type. References are matched, not resampled: for N and R the twin gets the *same realized* sequence its treated partner received, index-aligned on the observer grid; for invert, $b_t^* = 1 - s_t$ on the twin itself. Untreated worlds "move toward" these references *more* than treated worlds move toward their actual broadcasts: passive medians 0.0119, 0.0300, 0.0073. Every ΔP is therefore negative: -0.0036 (F), -0.0058 (R), -0.0055 (N), each $p < 2 \times 10^{-5}$. Between-condition contrasts of ΔP are significant in neither campaign (F vs N: harsh $p = 0.084$, flagship $p = 0.63$; R vs N: $p = 0.95$). Raw P is thus dominated by the intrinsic dynamics above, which align best with realistic references and worst with noise (excluding the shock transition shifts every median by < 0.0004). Response activation cannot explain the hierarchy either: R is the most quiescent of the three ($\langle |b_k - s_k| \rangle$ 0.10, against 0.64 F and 0.36 N) yet has the largest raw pull.

Under the primary corrective regime Eq. 2 thus provides *no evidence of aggregate content-specific attraction beyond intrinsic dynamics.* The small negative ΔP, similar across signal types, is consistent with broadcasts mildly dampening that drift. This is the paper's methodological result: a statistic natural in empirical performativity studies registers large, internally consistent, entirely artifactual "steering" unless checked against passive twins. We offer ΔP as a paired diagnostic estimator for designs with an untreated twin, not a general identification theorem.

Structural spillovers, by contrast, are real. Agents persistently told the network is fragmenting over-connect it: mean degree 19.8 vs 12.3 in matched flagship runs at $t = 2000$ (paired Δmedian +7.2, $p \approx 7 \times 10^{-10}$). Post-shock fragmentation falls with it (flagship paired Δmedian -0.017, $r = -0.52$, $p \approx 10^{-3}$; directionally consistent but ns under harsh shock). False and noise broadcasts densify the network, and also flatten it. In the baseline ecology, relative to A (paired, harsh shock, per-run medians over $t \geq 500$), mean degree rises +1.1 under truth ($r = 0.49$), +3.8 under the inverted lie ($r = 0.80$) and +2.8 under noise ($r = 0.72$). On the same runs and window, Freeman centralization falls

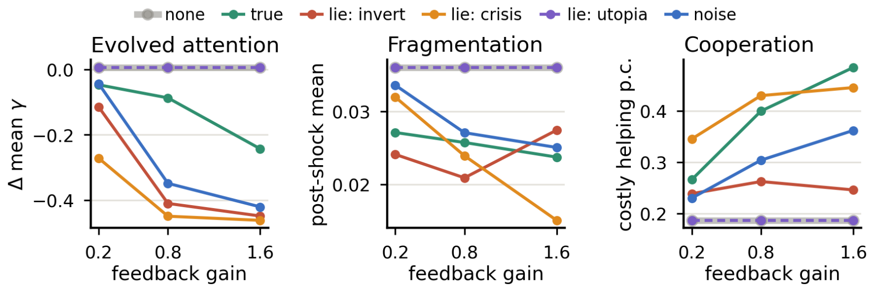

Fig. 2. Sensitivity sweep (40% hub-removal shock; 20 seeds per cell; medians). Left: attention varies strongly with broadcast type and gain, and the flattering (utopia) lie tracks the no-feedback baseline exactly (dashed over grey). Middle: under corrective responses the alarmist (crisis) lie does *not* come true. Right: deficit signals mobilize costly helping.

under both distortions (F: -0.026, $r = -0.74$; N: -0.010), as does betweenness concentration (F: -0.007, $r = -0.67$; N: -0.008).

5.3 The type of lie sets what changes

The three distortion modes produce qualitatively different *outcome* regimes (Fig. 2). The *inverted* self-model, always reporting the mirror of reality, keeps four of five corrective responses active on $\geq 90\%$ of ticks: it mobilizes costly helping (+0.106 over baseline, flagship $r = 0.82$) and drives the over-connection above. The *crisis* self-model is self-defeating for structure: corrections push fragmentation away from the alarmist report. Yet its permanent cooperation-deficit signal mobilizes the highest costly-helping rates of any lie under correction (0.35–0.45 across gains; only truthful feedback at $g = 1.6$ exceeds them, 0.48). The *utopia* self-model is behaviorally inert: responses fire only on reported deficits, so a flattering broadcast triggers nothing and is indistinguishable from disconnecting the channel at every gain. Under purely corrective rules, flattery equals silence. Caveat: these response *directions* follow from Eq. 1 by construction; the magnitudes, spillovers and evolutionary consequences below do not.

5.4 The response regime decides which lies come true

Conformist agents reverse the taxonomy exactly as the mechanism predicts (Table 4). The *crisis* lie, self-defeating under correction, becomes strongly self-fulfilling under conformity: told the world is fragmenting, imitators cut ties and fragmentation rises an order of magnitude above baseline (0.400 vs 0.038). The *utopia* lie flips from behaviorally inert to benevolently self-fulfilling: "everyone cooperates" becomes true by imitation (cooperation 0.428). Even noise mobilizes: imitating randomly reported cooperation levels raises costly helping to 0.445, as high as any lie, since imitation amplifies whatever is reported. Which false self-models come true — in the outcome sense — is therefore a property of the lie–response pair, not of the lie alone. Compactly: with macrostate update

Table 4. The same lie under two response regimes (medians over the 15 seeds shared by both campaigns; harsh shock, $g = 0.8$). Baseline (A), identical in both regimes: cooperation 0.183, fragmentation 0.038.

Lie	Corrective		Conformist	
	coop.	fragm.	coop.	fragm.
Invert	0.256	0.020	0.419	0.054
Crisis	0.414	0.018	0.255	**0.400**
Utopia	0.183 (inert)	0.038 (inert)	**0.428**	0.025

$S_{t+1} = \Phi\big(S_t, \mathcal{R}(\mathcal{B}(S_t))\big)$ for broadcast map $\mathcal{B}$ and response rule $\mathcal{R}$, self-fulfillment is a property of the composition $\mathcal{R} \circ \mathcal{B}$, not of $\mathcal{B}$ alone.

5.5 Attention declines under consequential broadcasts

Mean heritable global sensitivity $\bar{\gamma}$ changes over $\sim$20 generations of lineage depth as an emergent, unprogrammed function of broadcast quality. At $g = 0.8$ (harsh shock): no broadcast -0.03 (neutral drift), truth -0.10, noise -0.34 ($r = -0.94$, $p \approx 2 \times 10^{-7}$), false -0.42 (Δmedian -0.336 $[-0.456, -0.281]$, $r = -1.00$, $p \approx 1.9 \times 10^{-9}$). The sweep is dose-ordered: the decline under the inverted broadcast deepens from -0.12 ($g = 0.2$) to -0.45 ($g = 1.6$), and at the highest gain even *accurate* feedback declines markedly (-0.24). The categorical ordering compresses into one continuous relationship. Define the broadcast-implied corrective-drive intensity $I = g\,\langle \sum_a f_a(b(t)) \rangle$, computable from the broadcast alone. Evolved attention tracks I across the 15 sweep cells almost perfectly: Spearman $\rho = -0.99$, and $\rho = -0.83$ at run level ($n = 300$). Utopia ($I = 0$) is untouched, truth declines less where its corrective drive is smaller at equilibrium, and high-drive broadcasts pay most: corrective-drive intensity, rather than veracity alone, appears to organize the response.

Two caveats. I omits the per-agent factor γ_i, the quantity under selection, and comes from realized rather than independently randomized trajectories; we treat it as descriptive, not causal mediation. And it is compared only within corrective-regime sweeps, where the same f_a apply — a within-regime ordering, not one normalized across regimes.

Not an architectural artifact. Two asymmetries in the response architecture could in principle manufacture this pattern. First, feedback adds weight only to non-reproductive actions. Write $D(b) = \sum_a f_a(b)$ for that added mass, whose gain-scaled mean is I, and let W_0 be the base total weight. Then $P(\text{reproduce} \,|\, \gamma_i, b) = w_r/(W_0 + \gamma_i g D(b))$ falls with γ_i whenever $D(b) > 0$. Second, hub-targeted pruning fires with probability γ_i rather than $\gamma_i g$, a second, differently scaled channel. We reran all five broadcast conditions against each, 300 runs apiece. The reproduction-neutral variant rescales w_r by $(1 + \gamma_i g D/N_0)$ (N_0 the base non-reproductive mass), holding the selection probability at its no-broadcast value with other feedback weights unchanged; the second fixes hub-pruning probability at 0.5 for every agent.

The pattern survives both controls. Taken over the cells with a negative baseline $\Delta\bar{\gamma}$, median retention is 98% under the reproduction-neutral control and 100% under the γ-free-pruning one. The intensity correlation is almost unchanged ($\rho = -0.98$ and -1.00). Under the reproduction control, truth in fact declines *more* at $g \geq 0.8$. One of the fifteen γ-free-pruning cells shows an uncorrected shift: noise at $g = 0.8$ attenuates to 77% of its baseline decline (paired $p = 0.033$). The intensity ordering itself tightens rather than loosens. Selection on γ is ecological rather than an artifact of either channel, and the decline is not specific to the corrective rule form: under the conformist mapping $\bar{\gamma}$ still falls in every broadcast condition (paired Δmedian -0.14 to -0.57, all raw $p \leq 0.022$), though neither that ordering nor I carries over ($\rho = +0.10$, $n = 5$), as the scope above implies.

Evolvable response strength. With f_a fixed, lowering γ is the population's only route to disengagement. Making the polarity and strength $\rho_a \in [-2, 2]$ of each action channel heritable — $\rho_a = 1$ reproduces the corrective rule bit-identically, 0 ignores that channel, <0 reverses it — opens a second route, and $\bar{\gamma}$ still declines (vs. A, $n = 100$: C -0.125, F -0.222, N -0.212; all raw $p < 10^{-4}$). On matched seeds this attenuates the decline under false and noise feedback ($+0.126$, $r = 0.66$; $+0.165$, $r = 0.65$; both raw $p \approx 0.001$) but not under truth ($p = 0.90$): populations use both routes exactly where disengagement pays. The evolutionary result is therefore not an artifact of making γ the sole adjustable response; the individual-level pathway remains open.

5.6 Truthful feedback doubles cooperation; main-campaign recovery is unchanged

True feedback roughly doubles costly helping (flagship: 0.397 vs 0.179; paired Δmedian $+0.200$ $[0.172, 0.239]$, $r = 1.00$, $p \approx 9 \times 10^{-15}$; replicated under harsh shock, $r = 0.99$), rising monotonically with gain ($0.27 \rightarrow 0.48$). The loop implements distributed corrective feedback: deficits reported at the population level are repaired by uncoordinated individual action. By contrast, a pre-specified null: recovery time showed no feedback effect in either main campaign, in any condition, even with 40% of hubs removed. Every paired Δmedian is 0 and every $p > 0.22$ (baseline medians 10 and 40 steps). One lead runs the other way: at $2\times$ scale the inverted lie delayed recovery in every non-tied pair (+10 steps, $r = 1.00$), though with only five non-tied pairs 0.0625 is the smallest p attainable and it does not replicate at base scale. Under this recovery criterion, resilience appears predominantly demographic rather than informational.

5.7 Robustness to system scale and to the baseline ecology

Re-running the core conditions (A, C, $\mathrm{F}_{\mathrm{invert}}$, N) at doubled population and resources (steady-state $\approx$ 110–130 vs $\approx$ 60; $n = 8$) preserves the main effects: cooperation under truth 0.391 vs 0.173 (Δmedian $+0.151$ $[0.119, 0.247]$, $r = 1.00$, $p = 0.008$) and the attention ordering (F vs A -0.299 $[-0.388, -0.212]$, $r = -1.00$, $p = 0.008$; N vs A -0.219, $r = -0.89$, $p = 0.023$).

Ecological robustness. Perturbing six ecological parameters by ±25% gives 24 alternative ecologies, all passing viability criteria that were fixed and evaluated on the *no-broadcast* condition before any treatment ran. Across A/B/C/F/N × 10 paired seeds each (1,200 runs): cooperation rose under truth in 24/24, evolved attention fell under the inverted lie and under noise in 24/24, measurement without broadcast stayed exactly inert in every paired run, and F≤N≤C≤A held exactly in 21/24. The recovery null persisted; truthful densification did not (16/24 correct-signed), motivating the narrower claim in Sect. 5.2. Read as replication, this is *local* ecological robustness, not arbitrary-ecology generality.

6 Discussion and Limitations

First-order response directions are encoded in the rules; the findings live one level up. First, the methodological lesson: a statistic that looked like compelling evidence of self-fulfilling dynamics — positive, replicated, largest for realistic signals — was fully explained by intrinsic dynamics on passive twins. Empirical tests face this identification problem [22], typically without the untreated twin our platform provides. Second, the substantive findings: broadcasts change behavior and structure with a regime-dependent taxonomy of lies, and above all the evolutionary layer, where the credibility of a collective self-model becomes an evolving property. The response mappings are controlled mechanism probes, not calibrated behavioral models: they isolate how alternative receiver transformations turn identical broadcasts into different outcomes, and although their strength and polarity are now evolvable (Sect. 5.5) the underlying basis stays mechanistic; whether fully learned policies preserve the taxonomy is open. Further limitations: local rather than arbitrary-ecology generality (Sect. 5.7), and one network per run rather than interacting populations.

Metric-driven infrastructure — autoscalers, observability pipelines, ranking systems [4] — is this loop, and under deficit-driven responses *flattering telemetry can be operationally equivalent to no telemetry* while raising no alarms; agent populations sharing dashboards [15] are a natural empirical target, given an explicit passive baseline before any alignment statistic is read as steering.

7 Conclusion

We introduced a reproducible causal laboratory for reflexive self-modeling in adaptive networks and showed that (i) the measurement-only control is trajectory-identical to baseline, while broadcasting changes behavior, structure, and evolution; (ii) apparent "story pull" vanishes against a passive counterfactual, a caution for empirical performativity claims; (iii) which lies come true, in the outcome sense, is set by the lie–response pair, not the lie alone; and (iv) heritable attention to the collective self-model declines across generations, tracking broadcast-implied corrective drive within the corrective regime — reliance on a self-model is an evolving quantity that high-drive feedback spends down. Future work: learned response policies and language-model agents.

Reproducibility. Code, configurations, analysis scripts, pinned dependencies, and deterministic regeneration of every run: `github.com/Ivan825/Project-ONE`, tag `cn2026-submission`.

References

1. Gross, T., Blasius, B.: Adaptive coevolutionary networks: a review. J. R. Soc. Interface 5, 259–271 (2008)
2. Holme, P., Saramäki, J.: Temporal networks. Phys. Rep. 519, 97–125 (2012)
3. Soros, G.: Fallibility, reflexivity, and the human uncertainty principle. J. Econ. Methodol. 20, 309–329 (2013)
4. Espeland, W.N., Sauder, M.: Rankings and reactivity: how public measures recreate social worlds. Am. J. Sociol. 113, 1–40 (2007)
5. Merton, R.K.: The self-fulfilling prophecy. Antioch Rev. 8, 193–210 (1948)
6. Strathern, M.: 'Improving ratings': audit in the British university system. Eur. Rev. 5, 305–321 (1997)
7. Albert, R., Jeong, H., Barabási, A.-L.: Error and attack tolerance of complex networks. Nature 406, 378–382 (2000)
8. Kuramoto, Y.: Chemical Oscillations, Waves, and Turbulence. Springer (1984)
9. Skyrms, B.: Signals: Evolution, Learning, and Information. Oxford Univ. Press (2010)
10. Epstein, J.M., Axtell, R.: Growing Artificial Societies. Brookings/MIT Press (1996)
11. Bonabeau, E.: Agent-based modeling: methods and techniques for simulating human systems. PNAS 99, 7280–7287 (2002)
12. Kazil, J., Masad, D., Crooks, A.: Utilizing Python for agent-based modeling: the Mesa framework. In: SBP-BRiMS. Springer (2020)
13. Hagberg, A., Schult, D., Swart, P.: Exploring network structure, dynamics, and function using NetworkX. In: SciPy (2008)
14. Foerster, J., Assael, Y.M., de Freitas, N., Whiteson, S.: Learning to communicate with deep multi-agent reinforcement learning. In: NeurIPS (2016)
15. Park, J.S., O'Brien, J., Cai, C.J., Morris, M.R., Liang, P., Bernstein, M.S.: Generative agents: interactive simulacra of human behavior. In: UIST (2023)
16. Castellano, C., Fortunato, S., Loreto, V.: Statistical physics of social dynamics. Rev. Mod. Phys. 81, 591–646 (2009)
17. Perc, M., Jordan, J.J., Rand, D.G., Wang, Z., Boccaletti, S., Szolnoki, A.: Statistical physics of human cooperation. Phys. Rep. 687, 1–51 (2017)
18. Freeman, L.C.: Centrality in social networks: conceptual clarification. Soc. Netw. 1, 215–239 (1978)
19. Perdomo, J.C., Zrnic, T., Mendler-Dünner, C., Hardt, M.: Performative prediction. In: ICML. PMLR 119 (2020)
20. Wang, X., Yau, C.-Y., Wai, H.-T.: Network effects in performative prediction games. In: ICML. PMLR 202 (2023)
21. Góis, A., Mofakhami, M., Santos, F.P., Gidel, G., Lacoste-Julien, S.: Performative prediction on games and mechanism design. In: AISTATS. PMLR 258, 1855–1863 (2025)
22. Cheng, G., Hardt, M., Mendler-Dünner, C.: Causal inference out of control: estimating performativity without randomization. In: ICML. PMLR 235, 8077–8103 (2024)
23. Hardt, M., Mendler-Dünner, C.: Performative prediction: past and future. Stat. Sci. 40, 417–436 (2025)
24. Iwase, T., Tadokoro, Y., Fukuda, D.: Self-fulfilling signal of an endogenous state in network congestion games. Netw. Spat. Econ. 17, 889–909 (2017)